

(A)

$$01. \lim_{x \rightarrow 1} \frac{x^2 - 8x + 7}{7x^2 - 6x - 1}$$

$$\lim_{x \rightarrow 1} \frac{(x-7)(x-1)}{(7x+1)(x-1)}$$

$$\lim_{x \rightarrow 1} \frac{(x-7)}{7x+1}$$

$$\frac{-6}{8} = \frac{-3}{4}$$

$$7x^2 - 6x - 1$$

$$7x^2 - 7x + x - 1$$

$$7x(x-1) + (x-1)$$

$$(7x+1)(x-1)$$

$$x^2 - 8x + 7$$

$$x^2 - 7x - x + 7$$

$$x(x-7) - (x-7)$$

$$(x-7)(x-1)$$

$$02. \lim_{x \rightarrow 0} \frac{2x^2 - 5x^3 + 7x^4}{5x - 4x^2 - 5x^3}$$

$$\lim_{x \rightarrow 0} \frac{x^2(2 - 5x + 7x^2)}{x(5 - 4x - 5x^2)}$$

$$\frac{0}{0}$$

$$03. \lim_{x \rightarrow -2} \frac{x^2 + 3x + 2}{x^2 - x - 6}$$

$$\lim_{x \rightarrow -2} \frac{x(x^2 + 3x + 2)}{x^2 - x - 6}$$

$$\lim_{x \rightarrow -2} \frac{x(x+2)(x+1)}{(x-3)(x+2)}$$

$$\lim_{x \rightarrow -2} \frac{x(x+1)}{x-3}$$

$$\frac{(-2)(-2+1)}{-2-3}$$

$$= \frac{2}{5}$$

$$04. \lim_{x \rightarrow \frac{1}{2}} \frac{8x^3 - 1}{6x^2 - 5x + 1}$$

$$\lim_{x \rightarrow \frac{1}{2}} \frac{(2x)^3 - 1}{(3x-1)(2x-1)}$$

$$\lim_{x \rightarrow \frac{1}{2}} \frac{(2x-1)(4x^2 + 2x + 1)}{(3x-1)(2x-1)}$$

$$\frac{1+1+1}{\frac{3}{2}-1}$$

$$\frac{3}{\frac{1}{2}}$$

$$6$$

$$6x^2 - 5x + 1$$

$$6x^2 - 3x - 2x + 1$$

$$3x(2x-1) - (2x-1)$$

$$(3x-1)(2x-1)$$

SchoolMate

$$05. \lim_{x \rightarrow 1} \frac{x^2 - 3x + 2}{x^2 - 4x + 3}$$

$$\lim_{x \rightarrow 1} \frac{(x-2)(x-1)}{(x-3)(x-1)}$$

$$\frac{1}{2}$$

$$x^3 + x - 2 = (x-1)(Ax^2 + Bx + C)$$

$$x^3 + x - 2 = Ax^3 + Bx^2 + Cx - Ax^2 - Bx - C$$

සමාන පදો සමාන,

$$x^3 \Rightarrow \boxed{A=1}$$

$$x^0 \Rightarrow \boxed{C=2}$$

$$x^2 \Rightarrow \boxed{B=1}$$

$$06. \lim_{x \rightarrow 1} \frac{x^3 + x - 2}{x^3 - x^2 - x + 1}$$

$$\lim_{x \rightarrow 1} \frac{(x-1)(x^2 + x + 2)}{x^2(x-1) - (x-1)}$$

$$\lim_{x \rightarrow 1} \frac{(x-1)(x^2 + x + 2)}{(x^2 - 1)(x-1)}$$

$$\frac{1+1+2}{1-1}$$

$\frac{0}{0}$

$$x^3 + x - 2 = (x-1)(x^2 + x + 2)$$

$$07. \lim_{x \rightarrow 1} \frac{x^2 - 2x + 1}{x^3 - x}$$

$$\lim_{x \rightarrow 1} \frac{(x-1)^2}{x(x^2-1)}$$

$$\lim_{x \rightarrow 1} \frac{(x-1)^2}{x(x-1)(x+1)}$$

$\frac{0}{0}$

$$08. \lim_{x \rightarrow 1} \left(\frac{1}{1-x} - \frac{3}{1-x^3} \right)$$

$$\lim_{x \rightarrow 1} \left[\frac{1}{(1-x)} - \frac{3}{(1-x)(1+x+x^2)} \right]$$

$$\lim_{x \rightarrow 1} \frac{1+x+x^2-3}{(1-x)(1+x+x^2)}$$

$$(-) \lim_{x \rightarrow 1} \frac{x^2 + x - 2}{(x-1)(x^2 + x + 1)}$$

$$(-) \lim_{x \rightarrow 1} \frac{(x+2)(x-1)}{(x-1)(x^2 + x + 1)}$$

$$= \frac{3}{3} = 1$$

$$09. \lim_{x \rightarrow 2} \left[\frac{1}{x(x-2)^2} - \frac{1}{(x^2-3x+2)} \right]$$

$$\lim_{x \rightarrow 2} \left[\frac{1}{x(x-2)^2} - \frac{1}{(x-2)(x-1)} \right]$$

$$\lim_{x \rightarrow 2} \frac{(x-1) - x(x-2)}{x(x-2)^2(x-1)}$$

$$\lim_{x \rightarrow 2} \frac{x-1 - x^2+2x}{x(x-2)^2(x-1)}$$

$$\lim_{x \rightarrow 2} \frac{-x^2+2x-1}{x(x-2)^2(x-1)}$$

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(B)

$$01. \lim_{x \rightarrow 0} \frac{\sqrt{1-x} - 1}{x}$$

$$\lim_{x \rightarrow 0} \frac{(\sqrt{1-x} - 1)(\sqrt{1-x} + 1)}{x(\sqrt{1-x} + 1)}$$

$$\lim_{x \rightarrow 0} \frac{1-x-1}{x(\sqrt{1-x} + 1)}$$

$$(-) \lim_{x \rightarrow 0} \left(\frac{1}{\sqrt{1-x} + 1} \right)$$

$$-\frac{1}{2}$$

$$03. \lim_{x \rightarrow 5} \frac{\sqrt{x-1} - 2}{x-5}$$

$$\lim_{x \rightarrow 5} \frac{(\sqrt{x-1} - 2)(\sqrt{x-1} + 2)}{(x-5)(\sqrt{x-1} + 2)}$$

$$\lim_{x \rightarrow 5} \frac{x-1-4}{(x-5)(\sqrt{x-1} + 2)}$$

$$= \frac{1}{4}$$

SchoolMate

$$10. \lim_{x \rightarrow 2} \frac{x^3 - 3x^2 + 4}{x^4 - 8x^2 + 16}$$

$$\lim_{x \rightarrow 2} \frac{(x-2)(x^2-9x+2)}{(x^2-4)^2}$$

$$\lim_{x \rightarrow 2} \frac{(x-2)(x-2)(x+1)}{(x-2)(x+2)^2}$$

$$\lim_{x \rightarrow 2} \frac{x+1}{(x+2)^2}$$

$$\frac{3}{16}$$

$$02. \lim_{x \rightarrow 0} \frac{x}{\sqrt{1+x} - 1}$$

$$\lim_{x \rightarrow 0} \frac{x(\sqrt{1+x} + 1)}{(\sqrt{1+x} - 1)(\sqrt{1+x} + 1)}$$

$$\lim_{x \rightarrow 0} \frac{x(\sqrt{1+x} + 1)}{1+x-1}$$

$$2$$

$$04. \lim_{x \rightarrow 0} \frac{\sqrt{x^2+1} - 1}{\sqrt{x^2+16} - 4}$$

$$\lim_{x \rightarrow 0} \frac{(\sqrt{x^2+1} - 1)(\sqrt{x^2+1} + 1)(\sqrt{x^2+16} + 4)}{(\sqrt{x^2+16} - 4)(\sqrt{x^2+16} + 4)(\sqrt{x^2+1} + 1)}$$

$$\lim_{x \rightarrow 0} \frac{(x^2+1-1)(\sqrt{x^2+16} + 4)}{(x^2+16-16)(\sqrt{x^2+1} + 1)}$$

$$\frac{8}{2} = 4$$

$$\text{as. } \lim_{x \rightarrow 0} \frac{\sqrt[3]{1+x^2} - 1}{x^2}$$

$$\lim_{x \rightarrow 0} \frac{(1+x^2)^{1/3} - 1}{x^2}$$

$$\lim_{x \rightarrow 0} \frac{(1+x^2)^{1/3} - 1}{1+x^2 - 1}$$

$$\lim_{x \rightarrow 0} \left(\frac{(1+x^2)^{1/3} - 1}{[(1+x^2)^{1/3}]^2 - 1} \right)$$

$$\lim_{x \rightarrow 0} \frac{(1+x^2)^{1/3} - 1}{[(1+x^2)^{1/3} - 1][(1+x^2)^{1/3} + (1+x^2)^{1/3} + 1]}$$

$$\lim_{x \rightarrow 0} \frac{1}{(1+x^2)^{1/3} + (1+x^2)^{1/3} + 1}$$

$$\frac{1}{3}$$

$$\text{ob. } \lim_{x \rightarrow 1} \frac{x^2 - \sqrt{x}}{\sqrt{x} - 1}$$

$$\lim_{x \rightarrow 1} \frac{(x^2 - \sqrt{x})(x^2 + \sqrt{x})(\sqrt{x} + 1)}{(\sqrt{x} - 1)(\sqrt{x} + 1)\cancel{\sqrt{x}(x^2 + \sqrt{x})}}$$

$$\lim_{x \rightarrow 1} \frac{(x^4 - x)(\sqrt{x} + 1)}{(x - 1)(x^2 + \sqrt{x})}$$

$$\lim_{x \rightarrow 1} \frac{x(x^3 - 1)(\sqrt{x} + 1)}{(x - 1)(x^2 + \sqrt{x})}$$

$$\lim_{x \rightarrow 1} \frac{x(x - 1)(x^2 + x + 1)(\sqrt{x} + 1)}{(x - 1)(x^2 + \sqrt{x})}$$

$$\frac{1 \cdot (1 + 1 + 1)(1 + 1)}{1 + 1}$$

$$3$$

(C)

$$01. \lim_{n \rightarrow \infty} \frac{n+1}{n}$$

$$\lim_{n \rightarrow \infty} \frac{n(1 + 1/n)}{n}$$

$$\frac{1 + \frac{1}{\infty}}{1}$$

$$03. \lim_{n \rightarrow \infty} \frac{2n^2 + 3n - 1}{n^2 - 2n + 3}$$

$$\lim_{n \rightarrow \infty} \frac{n^2(2 + 3/n - 1/n^2)}{n^2(1 - 2/n + 3/n^2)}$$

$$\frac{2}{1}$$

$$05. \lim_{n \rightarrow \infty} \frac{\sqrt{2n^2 + 1}}{2n - 1}$$

$$\lim_{n \rightarrow \infty} \frac{n\sqrt{2 + 1/n^2}}{n(2 - 1/n)}$$

$$\frac{\frac{\sqrt{2}}{2}}{2} = \frac{1}{\sqrt{2}}$$

$$07. \lim_{n \rightarrow \infty} \left(\frac{1+2+3+\dots+n}{n^2} \right)$$

$$\lim_{n \rightarrow \infty} \frac{n(n+1)/2}{n^2}$$

$$\lim_{n \rightarrow \infty} \frac{n^2(1 + 1/n)}{2n^2}$$

$$\frac{1}{2}$$

$$02. \lim_{n \rightarrow \infty} \frac{(n+1)^2}{2n^2}$$

$$\lim_{n \rightarrow \infty} \frac{n^2(1 + 1/n)^2}{2n^2}$$

$$\frac{(1 + \frac{1}{\infty})^2}{2}$$

$$\frac{1}{2}$$

$$04. \lim_{n \rightarrow \infty} \frac{n^3 - 100n^2 + 1}{100n^2 + 15n}$$

$$\lim_{n \rightarrow \infty} \frac{n^3(1 - 100/n + 1/n^3)}{n^2(100 + 15/n)}$$

$$\frac{\infty}{\infty}$$

$$06. \lim_{n \rightarrow \infty} \frac{\sqrt[3]{n^3 + 2n - 1}}{n + 2}$$

$$\lim_{n \rightarrow \infty} \frac{n\sqrt[3]{1 + 2/n^2 - 1/n^3}}{n(1 + 2/n)}$$

$$\frac{1}{1}$$

$$1 + 2 + 3 + \dots + n = \frac{n(n+1)}{2}$$

$$1^2 + 2^2 + 3^2 + \dots + n^2 = \frac{n(n+1)(2n+1)}{6}$$

$$1^3 + 2^3 + 3^3 + \dots + n^3 = \frac{n^2(n+1)^2}{4}$$

$$8. \lim_{x \rightarrow \infty} \frac{(2x - \sqrt{4x^2 + x})}{(2x - \sqrt{4x^2 + x})(2x + \sqrt{4x^2 + x})}$$

$$\lim_{x \rightarrow \infty} \frac{4x^2 - 4x^2 - x}{2x + \sqrt{4x^2 + x}}$$

$$\lim_{x \rightarrow \infty} \frac{-x}{2x + x\sqrt{4 + \frac{1}{x}}}$$

$$= \frac{-1}{2+2}$$

$$= \underline{\underline{-\frac{1}{4}}}$$

$$\boxed{\lim_{n \rightarrow a} \frac{n^n - a^n}{n - a} = n \cdot a^{n-1}}$$

(9)
01. $\lim_{x \rightarrow 5} \frac{x^4 - 625}{x - 5}$

$$\lim_{x \rightarrow 5} \frac{x^4 - 5^4}{x - 5}$$

$$4(5)^3$$

$$4 \times 125$$

$$500$$

$$\underline{\underline{500}}$$

02. $\lim_{x \rightarrow 2} \frac{x^5 - 32}{x - 2}$

$$\lim_{x \rightarrow 2} \frac{x^5 - 2^5}{x - 2}$$

$$5(2)^4$$

$$5 \times 16$$

$$80$$

$$\underline{\underline{80}}$$

03. $\lim_{x \rightarrow 2} \frac{x^{10} - 1024}{x^5 - 32}$

$$\lim_{x \rightarrow 2} \frac{x^{10} - 2^{10}}{x^5 - 2^5}$$

$$\lim_{x \rightarrow 2} \frac{x^{10} - 2^{10}}{x^5 - 2^5} \cdot \frac{x-2}{x-2}$$

$$\frac{10(2)^9}{5(2)^4}$$

$$2(2)^5$$

$$64$$

$$\underline{\underline{64}}$$

$$04. \lim_{x \rightarrow -3} \frac{x^5 - 243}{x + 3}$$

$$\lim_{x \rightarrow -3} \frac{x^5 - (-3)^5}{x - (-3)}$$

$$5(-3)^4$$

$$5 \times 81$$

$$405$$

405

$$06. \lim_{x \rightarrow 3} \frac{x^5 - 243}{x^2 - 9}$$

$$\lim_{x \rightarrow 3} \frac{x^5 - 3^5}{x^2 - 3^2} \cdot \frac{x - 3}{x - 3}$$

$$5(3)^4$$

$$2(3)$$

$$5 \times 27$$

$$2$$

$$135$$

$$2$$

135

$$07. \lim_{x \rightarrow a} \frac{(x+2)^{5/3} - (a+2)^{5/3}}{x - a}$$

$$\lim_{x \rightarrow a} \frac{(x+2)^{5/3} - (a+2)^{5/3}}{x+2 - (a+2)}$$

$$x \rightarrow a \text{ so}$$

$$x+2 \rightarrow a+2$$

$$\lim_{x+2 \rightarrow a+2} \frac{(x+2)^{5/3} - (a+2)^{5/3}}{x+2 - (a+2)}$$

$$\frac{5}{3} (a+2)^{2/3}$$

$\frac{5}{3} (a+2)^{2/3}$

$$05. \lim_{x \rightarrow 0} \left(\frac{\sqrt{1+x} - 1}{\sqrt[3]{1+x} - 1} \right)$$

$$\lim_{x \rightarrow 0} \frac{(1+x)^{1/2} - 1^{1/2}}{(1+x)^{1/3} - 1^{1/3}}$$

$$\lim_{x \rightarrow 0} \frac{\left(\frac{(1+x)^{1/2} - 1^{1/2}}{(1+x) - 1} \right)}{\left(\frac{(1+x)^{1/3} - 1^{1/3}}{(1+x) - 1} \right)}$$

$$\cancel{x \rightarrow 0} \text{ so}$$

$$1+x \rightarrow 1$$

$$\lim_{1+x \rightarrow 1} \frac{\left(\frac{(1+x)^{1/2} - 1^{1/2}}{(1+x) - 1} \right)}{\left(\frac{(1+x)^{1/3} - 1^{1/3}}{(1+x) - 1} \right)}$$

$$\frac{1/2(1)^{-1/2}}{1/3(1)^{-2/3}}$$

$$\frac{3}{2}$$

$$2$$

3/2

$$08. \lim_{x \rightarrow 0} \frac{(1+x)^5 - 1}{3x + 5x^2}$$

$$\lim_{x \rightarrow 0} \frac{(1+x)^5 - 1}{x(5x+3)}$$

$$\lim_{x \rightarrow 0} \frac{(1+x)^5 - 1}{(1+x) - 1} \cdot \frac{1}{(5x+3)}$$

$$\lim_{x \rightarrow 0} \frac{(1+x)^5 - 1^5}{(1+x) - 1} \cdot \lim_{x \rightarrow 0} \frac{1}{5x+3}$$

$$x \rightarrow 0 \Rightarrow$$

$$1+x \rightarrow 1$$

$$\frac{1}{3} \lim_{x \rightarrow 1} \frac{(1+x)^5 - 1^5}{(1+x) - 1}$$

$$\frac{1}{3} \cdot 5(1)^4$$

$$\underline{\underline{5/3}}$$

$$10. \lim_{x \rightarrow 27} \frac{(x^{1/3} + 3)(x^{1/3} - 3)}{x - 27}$$

$$\lim_{x \rightarrow 27} \frac{x^{2/3} - 27^{2/3}}{x - 27}$$

$$\frac{2}{3} (27)^{-1/3}$$

$$\frac{2}{3} \times 3^{3(-1/3)}$$

$$\underline{\underline{\frac{2}{9}}}$$

$$09. \lim_{x \rightarrow a} \frac{x^{-3} - a^{-3}}{x^{-5} - a^{-5}}$$

$$\lim_{x \rightarrow a} \frac{x^{-3} - a^{-3}}{x^{-5} - a^{-5}} \cdot \frac{x-a}{x-a}$$

$$\frac{(-3) a^{-4}}{(-5) a^{-6}}$$

$$\underline{\underline{\frac{3a^2}{5}}}$$

$$9 = 27^{2/3}$$

$$3 = 3^{3 \times 2/3}$$

$$= 9$$

(E)

$$01. \lim_{x \rightarrow 0} \frac{\sin 3x}{5x}$$

$$\frac{1}{5} \lim_{x \rightarrow 0} \frac{\sin 3x}{3x} \cdot 3$$

$$\underline{\underline{3/5}}$$

$$02. \lim_{x \rightarrow 0} \frac{1 - \cos x}{x^2}$$

$$\lim_{x \rightarrow 0} \frac{2 \sin^2 x/2}{x^2}$$

$$\lim_{x \rightarrow 0} \frac{2 \sin^2 x/2}{\left(\frac{x}{2}\right)^2} \cdot \frac{1}{4}$$

$$\underline{\underline{1/2}}$$

$$03. \lim_{x \rightarrow 0} \frac{1 - \cos 2x}{x \cdot \sin x}$$

$$\lim_{x \rightarrow 0} \frac{2 \sin^2 x}{x \cdot \sin x}$$

$$\lim_{x \rightarrow 0} \frac{2 \sin^2 x}{x^2} \cdot \frac{x}{\sin x}$$

$$\underline{\underline{2}}$$

$$04. \lim_{x \rightarrow 0} \left(\frac{\sin 3x}{x} - \frac{\sin 2x}{x} \right)$$

$$\lim_{x \rightarrow 0} \frac{\sin 3x}{3x} \cdot 3 - \lim_{x \rightarrow 0} \frac{\sin 2x}{2x} \cdot 2$$

$$\underline{\underline{1}}$$

$$05. \lim_{x \rightarrow 0} \frac{\sin x^\circ}{x}$$

$$x^\circ \rightarrow x \cdot \frac{\pi}{180}$$

$$\lim_{x \rightarrow 0} \frac{\sin\left(x \cdot \frac{\pi}{180}\right)}{x}$$

$$\lim_{x \rightarrow 0} \frac{\sin x}{x} \cdot \frac{\pi}{180}$$

$$\underline{\underline{\pi/180}}$$

$$06. \lim_{x \rightarrow 0} \frac{\sin 2x (1 - \cos 2x)}{x^3}$$

$$\lim_{x \rightarrow 0} \frac{\sin 2x}{x} \cdot \frac{2 \sin^2 x}{x^2}$$

$$\lim_{x \rightarrow 0} \frac{\sin 2x}{2x} \cdot 2 \cdot \lim_{x \rightarrow 0} \frac{2 \sin^2 x}{x^2}$$

$$\underline{\underline{4}}$$

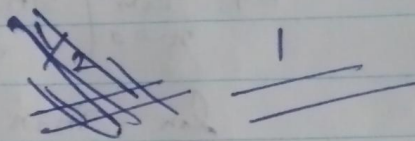
$$07. \lim_{x \rightarrow 0} \frac{2 \sin x - \sin 2x}{x^3}$$

$$\lim_{x \rightarrow 0} \frac{2 \sin x - 2 \sin x \cos x}{x^3}$$

$$\lim_{x \rightarrow 0} \frac{2 \sin x (1 - \cos x)}{x^3}$$

$$\lim_{x \rightarrow 0} \frac{2 \sin x}{x} \cdot \lim_{x \rightarrow 0} \frac{2 \sin^2 x/2}{x^2}$$

$$\lim_{x \rightarrow 0} 2 \sin x \cdot \lim_{x \rightarrow 0} \frac{2 \sin^2 x/2}{(x/2)^2} = \frac{1}{4}$$



$$08. \lim_{x \rightarrow 0} \frac{1 - \cos x}{1 - \cos 3x}$$

$$\lim_{x \rightarrow 0} \frac{2 \sin^2 x/2}{2 \sin^2 3x/2}$$

$$\lim_{x \rightarrow 0} \left(\frac{\frac{\sin^2 x/2}{(x/2)^2} \cdot \frac{1}{4}}{\frac{\sin^2 3x/2}{(3x/2)^2} \cdot \frac{9}{4}} \right)$$

$$\frac{1}{9}$$

$$09. \lim_{x \rightarrow 0} \left(\frac{\tan 2x - \sin 2x}{x^3} \right)$$

$$\lim_{x \rightarrow 0} \left(\frac{\frac{\sin 2x}{\cos 2x} - \sin 2x}{x^3} \right)$$

$$\lim_{x \rightarrow 0} \frac{\sin 2x - \sin 2x \cos 2x}{x^3 \cdot \cos 2x}$$

$$\lim_{x \rightarrow 0} \frac{\sin 2x (1 - \cos 2x)}{x^3 \cdot \cos 2x}$$

$$\lim_{x \rightarrow 0} \frac{2 \sin 2x}{2x} \cdot \lim_{x \rightarrow 0} \frac{2 \sin^2 x}{x^2} \cdot \frac{1}{\cos 2x}$$

$$2 \cdot 2 \cdot \frac{1}{1}$$

$$4$$

$$10. \lim_{x \rightarrow 0} \frac{\tan x - \sin x}{\sin 3x - 3\sin x}$$

$$\lim_{x \rightarrow 0} \frac{\frac{\sin x}{\cos x} - \sin x}{(3\sin x - 4\sin^3 x) - 3\sin x}$$

$$\lim_{x \rightarrow 0} \frac{\sin x - \sin x \cos x}{\cos x (-4\sin^2 x)}$$

$$-\frac{1}{4} \lim_{x \rightarrow 0} \frac{\sin x (1 - \cos x)}{\sin^3 x \cos x}$$

$$-\frac{1}{4} \lim_{x \rightarrow 0} \frac{2\sin^2 x/2}{\sin^2 x} \cdot \lim_{x \rightarrow 0} \frac{1}{\cos x}$$

$$-\frac{1}{2} \lim_{x \rightarrow 0} \frac{(\sin^2 x/2) x^2}{\sin^2 x / x^2} \quad (1)$$

$$-\frac{1}{2} \lim_{x \rightarrow 0} \frac{\sin^2 x/2}{(x/2)^2} \cdot \frac{1}{4}$$

$$-\frac{1}{8}$$

$$12. \lim_{x \rightarrow 0} \left(\frac{\sqrt{1+\sin x} - \sqrt{1-\sin x}}{x} \right)$$

$$\lim_{x \rightarrow 0} \frac{(\sqrt{1+\sin x} - \sqrt{1-\sin x})(\sqrt{1+\sin x} + \sqrt{1-\sin x})}{x(\sqrt{1+\sin x} + \sqrt{1-\sin x})}$$

$$\lim_{x \rightarrow 0} \frac{1 + \sin x - 1 + \sin x}{x(\sqrt{1+\sin x} + \sqrt{1-\sin x})}$$

$$\lim_{x \rightarrow 0} \frac{2\sin x}{x} \cdot \lim_{x \rightarrow 0} \frac{1}{(\sqrt{1+\sin x} + \sqrt{1-\sin x})}$$

$$2 \cdot \frac{1}{2} = 1$$

$$11. \lim_{x \rightarrow 0} \frac{\sin 2x - 3x}{4x - \sin 5x}$$

$$\lim_{x \rightarrow 0} \frac{\left(\frac{\sin 2x - 3x}{x} \right)}{\left(\frac{4x - \sin 5x}{x} \right)}$$

$$\lim_{x \rightarrow 0} \left(\frac{\frac{\sin 2x}{x} - 3}{4 - \frac{\sin 5x}{x}} \right)$$

$$\lim_{x \rightarrow 0} \frac{\frac{\sin 2x}{2x} \cdot 2 - 3}{4 - \frac{\sin 5x}{5} \cdot 5}$$

$$\frac{2 - 3}{4 - 5}$$

$$= 1$$

$$13. \lim_{x \rightarrow 0} (\operatorname{cosec} x - \cot x)$$

$$\lim_{x \rightarrow 0} \frac{1}{\sin x} - \frac{\cos x}{\sin x}$$

$$\lim_{x \rightarrow 0} \frac{1 - \cos x}{\sin x}$$

$$\lim_{x \rightarrow 0} \frac{2 \sin^2 x/2}{2 \sin x/2 \cos x/2}$$

$$\lim_{x \rightarrow 0} \tan x/2$$

$$0$$

$$15. \lim_{x \rightarrow 0} \frac{(1 - \cos x) \sqrt{\cos 2x}}{x^2}$$

$$\lim_{x \rightarrow 0} \frac{2 \sin^2 x/2}{x^2} \cdot \lim_{x \rightarrow 0} \sqrt{\cos 2x}$$

$$\lim_{x \rightarrow 0} \frac{2 \sin^2 x/2}{(x/2)^2} \cdot \frac{1}{4} (1)$$

$$\frac{1}{2}$$

$$14. \lim_{x \rightarrow 0} \left(\frac{\sqrt{5} - \sqrt{4 + \cos x}}{x \sin x} \right)$$

$$\lim_{x \rightarrow 0} \frac{(\sqrt{5} - \sqrt{4 + \cos x})(\sqrt{5} + \sqrt{4 + \cos x})}{x \sin x (\sqrt{5} + \sqrt{4 + \cos x})}$$

$$\lim_{x \rightarrow 0} \frac{5 - 4 - \cos x}{x \sin x (\sqrt{5} + \sqrt{4 + \cos x})}$$

$$\lim_{x \rightarrow 0} \frac{1 - \cos x}{x \sin x (\sqrt{5} + \sqrt{4 + \cos x})}$$

$$\lim_{x \rightarrow 0} \frac{2 \sin^2 x/2}{x \cdot 2 \sin x/2 \cos x/2} \cdot \lim_{x \rightarrow 0} \frac{1}{(\sqrt{5} + \sqrt{4 + \cos x})}$$

$$\frac{2 \cdot \frac{1}{2}}{\sqrt{5} + \sqrt{5}}$$

$$\lim_{x \rightarrow 0} \frac{\sin(x/2)}{x/2} \cdot \frac{1}{2} \cdot \frac{1}{2\sqrt{5}} \cdot \lim_{x \rightarrow 0} \frac{1}{\cos x/2}$$

$$\frac{1}{4\sqrt{5}}$$

$$16. \lim_{x \rightarrow 0} \frac{1 - \cos 2x + \tan^2 x}{x \tan x}$$

$$\lim_{x \rightarrow 0} \frac{1 - \cos 2x}{x \tan x} + \lim_{x \rightarrow 0} \frac{\tan^2 x}{x \tan x}$$

$$\lim_{x \rightarrow 0} \frac{2 \sin^2 x}{x \cdot \sin x / \cos x} + \lim_{x \rightarrow 0} \frac{\sin^2 x / \cos^2 x}{x}$$

$$2 \lim_{x \rightarrow 0} \frac{\sin x}{x} \cdot \lim_{x \rightarrow 0} \cos x + \lim_{x \rightarrow 0} \frac{\sin x}{x} \cdot \lim_{x \rightarrow 0} \frac{1}{\cos x}$$

$$2(1) + 1(1) = 3$$

$$\lim_{x \rightarrow \pi/2} \frac{\cos x}{\pi/2 - x}$$

$$\lim_{y \rightarrow 0} \frac{\cos(\pi/2 - y)}{y}$$

$$\lim_{y \rightarrow 0} \frac{\sin y}{y} = 1$$

$$2. \lim_{x \rightarrow \pi} \frac{\sin x}{\pi - x}$$

$$\lim_{y \rightarrow 0} \frac{\sin(\pi - y)}{y}$$

$$\lim_{y \rightarrow 0} \frac{\sin y}{y}$$

1

$$y = \pi/2 - x \text{ so so,}$$

$$x = \pi/2 - y$$

$$\begin{array}{l} x \rightarrow \pi/2 \text{ so,} \\ y \rightarrow 0 \end{array}$$

$$y = \pi - x,$$

$$x = \pi - y$$

$$x \rightarrow \pi \text{ so}$$

$$y \rightarrow 0$$

$$3. \lim_{x \rightarrow 2\pi} \frac{\sin x}{2\pi - x}$$

$$y = 2\pi - x$$

$$x = 2\pi - y$$

$$\lim_{y \rightarrow 0} \frac{\sin(2\pi - y)}{y}$$

$$x \rightarrow 2\pi \text{ s.t.}$$

$$y \rightarrow 0$$

$$\lim_{y \rightarrow 0} \frac{(-) \sin y}{y}$$

$$= -1$$

$$\underline{\underline{-1}}$$

$$4. \lim_{x \rightarrow \infty} x \cdot \sin \frac{1}{x}$$

$$y = \frac{1}{x}$$

$$x = \frac{1}{y}$$

$$\lim_{y \rightarrow 0} \frac{\sin y}{y}$$

$$= 1$$

$$\underline{\underline{1}}$$

$$17. \lim_{x \rightarrow \pi/4} \frac{\sin x - \cos x}{1 - \tan x}$$

$$\lim_{x \rightarrow \pi/4} \frac{\sin x - \cos x}{1 - \sin x / \cos x}$$

$$\lim_{x \rightarrow \pi/4} \frac{(\sin x - \cos x) \cos x}{\cos x - \sin x}$$

$$(-) \lim_{x \rightarrow \pi/4} \cos x$$

$$-\frac{1}{\sqrt{2}}$$

$$(E) \text{ s. } \lim_{x \rightarrow 0} \frac{\sin^{-1} x}{x}$$

$$y = \sin^{-1} x$$

$$x = \sin y$$

$$\lim_{y \rightarrow 0} \frac{y}{\sin y}$$

$$-\pi/2 \leq y \leq \pi/2$$

$$\lim_{y \rightarrow 0} \left(\frac{y}{\sin y \cdot y} \right)$$

$$x \rightarrow 0 \text{ so}$$

$$y \rightarrow 0$$

$$\frac{1}{1}$$

$$\underline{\underline{1}}$$

$$06. \lim_{x \rightarrow \pi/2} (\pi/2 - x) \tan x$$

$$y = \pi/2 - x$$

$$x = \pi/2 - y$$

$$\lim_{y \rightarrow 0} y \cdot \tan(\pi/2 - y)$$

$$x \rightarrow \pi/2 \text{ so}$$

$$y \rightarrow 0$$

$$\lim_{y \rightarrow 0} y \cdot \cot y$$

$$\lim_{y \rightarrow 0} y \cdot \frac{\cos y}{\sin y}$$

$$\lim_{y \rightarrow 0} \frac{y}{\sin y} \cdot \cos y$$

$$\underline{\underline{1}}$$

$$08. \lim_{x \rightarrow 1} (1-x) \tan\left(\frac{\pi x}{2}\right)$$

$$y = 1 - x$$

$$x = 1 - y$$

$$\lim_{y \rightarrow 0} y \cdot \tan\left[\frac{\pi}{2}(1-y)\right]$$

$$x \rightarrow 1 \text{ so}$$

$$y \rightarrow 0$$

$$\lim_{y \rightarrow 0} y \cdot \tan\left(\frac{\pi}{2} - \frac{\pi y}{2}\right)$$

$$\lim_{y \rightarrow 0} y \cdot \cot\left(\frac{\pi y}{2}\right)$$

$$\lim_{y \rightarrow 0} y \cdot \frac{\cos(\pi y/2)}{\sin(\pi y/2)}$$

$$\lim_{y \rightarrow 0} \frac{y \cdot \pi/2}{\sin(\pi y/2)} \cdot \frac{2}{\pi} \cdot \lim_{y \rightarrow 0} \cos\left(\frac{\pi y}{2}\right)$$

$$\underline{\underline{\frac{2}{\pi}}}$$

$$\therefore \lim_{\theta \rightarrow \pi/2} \frac{c + \theta}{\pi/2 - \theta}$$

$$y = \pi/2 - \theta$$

$$\theta = \pi/2 - y$$

$$\lim_{y \rightarrow 0} \frac{c + (\pi/2 - y)}{y}$$

$$\theta \rightarrow \pi/2 \text{ so}$$

$$y \rightarrow 0$$

$$\lim_{y \rightarrow 0} \frac{\tan y}{y}$$

$$\underline{\underline{1}}$$

9. $\lim_{x \rightarrow \pi/6} \frac{\sqrt{3} \sin x - \cos x}{x - \pi/6}$ ~~$\lim_{x \rightarrow \pi/6}$~~ $y = x - \pi/6$
 $x = y + \pi/6$

$$\frac{\sqrt{3} \sin(y + \pi/6) - \cos(y + \pi/6)}{y}$$

$x \rightarrow \pi/6$ so

$y \rightarrow 0$

~~$$\frac{\sqrt{3} \sin y \cos \pi/6 + \cos y \sin \pi/6 - (\cos y \cos \pi/6 - \sin y \sin \pi/6)}{y}$$~~

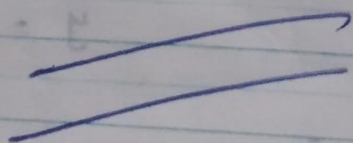
$$\frac{\sqrt{3} (\sin y \cos \pi/6 + \cos y \sin \pi/6) - (\cos y \cos \pi/6 - \sin y \sin \pi/6)}{y}$$

$$\frac{\sqrt{3} \left(\frac{\sqrt{3}}{2} \sin y + \frac{1}{2} \cos y \right) - \left(\frac{\sqrt{3}}{2} \cos y - \frac{1}{2} \sin y \right)}{y}$$

$$\frac{\frac{3}{2} \sin y + \frac{\sqrt{3}}{2} \cos y - \frac{\sqrt{3}}{2} \cos y + \frac{1}{2} \sin y}{y}$$

$$\frac{\sin y}{y}$$

≈ 1



$$10. \lim_{x \rightarrow \pi/4} \left(\frac{\sin x - \cos x}{x - \pi/4} \right)$$

$$y = x - \pi/4$$

$$x = y + \pi/4$$

$$\lim_{x \rightarrow \pi/4} \frac{\sin(y + \pi/4) - \cos(y + \pi/4)}{y}$$

$$x \rightarrow \pi/4 \text{ so } y = 0$$

$$\lim_{x \rightarrow \pi/4} \frac{\sin y \cos \pi/4 + \cos y \sin \pi/4 - (\cos y \cos \pi/4 - \sin y \sin \pi/4)}{y}$$

$$\lim_{x \rightarrow \pi/4} \frac{\frac{1}{\sqrt{2}} \sin y + \frac{1}{\sqrt{2}} \cos y - \frac{1}{\sqrt{2}} \cos y + \frac{1}{\sqrt{2}} \sin y}{y}$$

$$\sqrt{2} \lim_{x \rightarrow \pi/4} \frac{\sin y}{y}$$

$$\underline{\underline{\sqrt{2}}}$$

$$11. \lim_{x \rightarrow \pi} \frac{1 + \cos x}{(\pi - x)^2}$$

$$y = \pi - x$$

$$x \rightarrow \pi$$

$$x = \pi - y$$

$$y \rightarrow 0$$

$$\lim_{x \rightarrow \pi} \frac{1 + \cos(\pi - y)}{y^2}$$

$$\lim_{x \rightarrow \pi} \frac{1 - \cos y}{y^2}$$

$$\lim_{x \rightarrow \pi} \frac{2 \sin^2 y/2}{\left(\frac{y}{2}\right)^2} \cdot \frac{1}{4}$$

$$\underline{\underline{\frac{1}{2}}}$$